CHAPTER 3

TRAFFIC SOURCE MODELS

3.1 Introduction

The Information Flow model captures the average traffic generated by a source, but it overlooks the finer temporal dynamics. While this model is useful for rough network resource dimensioning or assessing system stability, it falls short when analyzing more specific performance metrics such as delay, traffic peaks, or for accurately sizing certain network components, such as buffers. For these purposes, it is necessary to describe the offered traffic profile, i.e., the pattern with which data are actually generated by the source, through suitable *traffic models*.

According to the final design goal, a traffic model can actually capture different aspects of the traffic generated by a source. For example, to have an idea of the order of magnitude of the bitrate required from a trunk connection, it suffices to model just the long-term average traffic offered by all the data sources multiplexed over that trunk. Instead, to check whether the average transmission delay of a certain Information Flow is below a given threshold, the model also needs to account for the time dynamics of the traffic source. In both cases, the actual content of the traffic is irrelevant for the network design and dimensioning, and hence it is usually not considered in traffic modeling. If the purpose is to design access schemes to support traffic prioritization, instead, the source model should also differentiate the packets generated by the source according to their priority class.

In the rest of this chapter, we will first illustrate a general method to describe a traffic source by means of mathematical models. Successively, we will discuss some basic traffic source models, which can be used to check the performance of some network manage-

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ment protocols (MAC and scheduling algorithms, in particular), or to provide a first, rough dimensioning of the network resources. Finally, we will present some methodologies to verify the model validity.

3.2 Abstractions of an Information Flow

An Information Flow is typically associated to a logical *session* between two (or more) End Nodes that need to exchange information. For example, when a web browser contacts a web server to get a certain World-Wide Web (WWW) content, the two End Nodes establish an Hypertext Transfer Protocol (HTTP) session. During the session, the nodes will actively exchange data in certain time intervals, and remain idle in others (e.g., when the user consults the content of the downloaded webpage). The session ends when the client connects to a different web server or it is closed by the user. A different example of a logical session between two End Nodes may be that of a mailer fetching new emails from a mail server. In this case, the session starts with the request from the client to the mail server, and ends when all the emails are downloaded to the mailer application. Therefore, traffic exchange during a session may be irregular, with alternating periods of active data exchange with periods of inactivity. During the active periods, furthermore, the traffic exchanged by the End Node can be constant or variable, according to the type of application.

Depending on the purpose of the traffic model, therefore, we can describe an Information Flow at different levels of abstractions, as depicted in Fig. 3.1, i.e.,

- *Session level*: ithe model only describes when an Information Flow starts and ends, and what is the average generated traffic. In other words, the model describes a logical *session* between two End Nodes;
- *Activity level*: the model describes the active and idle periods during a session, by defining the duration of such periods (usually, in statistical terms) and the average traffic generated in each period;
- *Packet level*: the model describes in details the packet generation pattern and packet size distribution during active and idle periods.

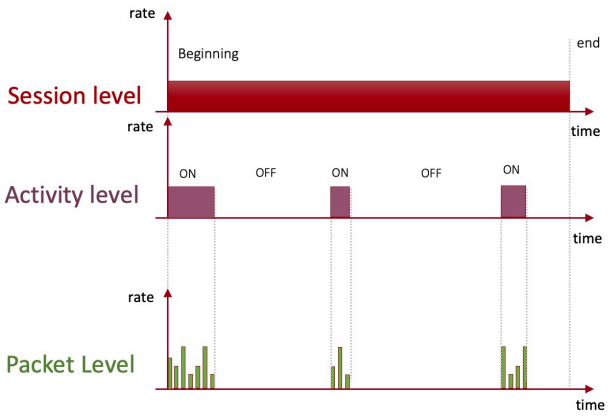


Figure 3.1 Graphical representation of the different levels of abstraction of a source model.

In the following, we first introduce a very general model that can be used to represent a large variety of traffic sources. Then, we will present some very common traffic source models that are often used in simulation or for mathematical analysis of communication systems.

3.3 A general model: Marked and Weighted Point Process

A Point Process (PP) is a stochastic model used to represent the occurrence of discrete random events in some continuous domain, typically the time axis or a Cartesian plane. In general, events (and their continuous domains) can be of the most diverse nature.

In the context of source modeling, the domain is typically the time axis, and the events are connected to the generation of new traffic. Therefore, depending on the level of abstraction of the source traffic model, events can correspond to the beginning or ending of a session, or to the beginning/ending of active/inactive periods, or to the generation of new packets. In any case, we will refer to any such events as an *arrival*. Furthermore, without loss of generality, we can describe the arrival process from time zero onward.

Mathematically, the PP can then be represented by the set $\mathcal{T} = \{\tau_i\}$ of arrival instants, where $\tau_i \in \mathcal{R}^+$ indicates the time at which the ith arrival occurs, with $i = 1, 2, \ldots$. An equivalent representation of the process can be given in terms of the so-called *inter-arrival times*, i.e., the set $\mathcal{A} = \{A_i\}$, where $A_i = \tau_i - \tau_{i-1}$ is the time between the $i-1$st and the ith arrival. A third, equivalent way to represent the PP is by means of the associated *counting process* $N(t) : \mathcal{R} \to \mathcal{N}$, which represents the number of arrivals of the PP in the time interval $(0, t]$, i.e.,

$$N(t) = \arg\max_n \{\tau_n \leq t\} = \sum_{k=0}^{\infty} \chi\{\tau_k \leq t\} \ . \tag{3.1}$$

For ease of notation, the number of arrivals occurring in a given time interval $(t_1, t_2]$, with $0 \leq t_1 \leq t_2$, will be indicated as $N(t_1, t_2) = N(t_2) - N(t_1)$.

A Point Process (PP) can be extended to include additional information associated with each event, resulting in a *marked point process* or a *weighted point process*, as explained next.

3.3.1 Marked Point Process

A generalization of the PP is the Marked Point Process, where each arrival event can be associated to a certain numerousness, indicated by a compound process $\mathcal{B} = \{B_i\}$, $i = 0, 1, 2, \ldots$, where $B_i \in \mathbb{N}_0$ is the *mark* associated to the ith arrival. Mathematically, the process can be represented as:

$$(\mathcal{T}, \mathcal{B}) \, , \quad \text{or equivalently} \quad (\mathcal{A}, \mathcal{B}) \, ,$$

where B_i is the mark associated to the ith arrival. The marked point process is called *independent* if the marks B_i are independent of the arrival times τ_i. Otherwise, it is said to be *dependent*.

The marks B_i can represent characteristics such as number of Information Units (IUs) generated at the ith arrival event, the type of the IUs, or the priority. In the following, we assume that the mark actually represents the *batch size*, i.e., the number of IU generated at each arrival event.

Consider, for example, an application that generates messages to be delivered to a certain destination. Such messages can consist of multiple packets that are passed to the network layer for transmission. Then, from the perspective of the network layer, the generation of a new message by the source can be modelled as an arrival that, however, brings along a bunch of packets to be transmitted, the numerousness of which is modelled by the process $\mathcal{B}$.

3.3.2 Marked and Weighted Point Process

A Marked and Weighted Point Process (MWPP) enriches the marked point process $(\mathcal{T}, \mathcal{B})$ with the *weight* process $\mathcal{W} = \{W_{i,k}\}$, which assigns a weight $W_{i,k}$ to each IU $k \in \{1, \ldots, B_i\}$ in the batch associated with arrival τ_i. The weight represents the significance or contribution of the IU for the system. For example, considering the kth IU in the ith arrival batch, $W_{i,k}$ can be the length in bits of the IU, or the energy consumption required for its transmission, or the computation required for its processing. Therefore, these weights are hence associated to a certain time of *workload*.

Therefore, a MWPP is a generalized representation of a PP that includes both marks and weights, and is hence defined by the triplet:

$$(\mathcal{T}, \mathcal{B}, \mathcal{W}) \quad \text{or equivalently} \quad (\mathcal{A}, \mathcal{B}, \mathcal{W})\,.$$

In this books, whenever the MWPP is used to model a packet generation process, the process $\mathcal{T}$ models the generation events, $\mathcal{B}$ indicates the batch sizes, i.e., the number of packets simultaneously generated at each arrival event,[1] and $\mathcal{W}$ represents the workload associated with each packet. In this case, the rate of the traffic generated by the application in a certain time interval $(t_1, t_2]$ can be expressed as

$$\mathrm{G}(t_1, t_2) = \frac{\sum_{i=N(t_1)+1}^{N(t_2)} \sum_{k=1}^{B_k} W_{i,k}}{t_2 - t_1}\,. \tag{3.2}$$

The MWPP offers a very general model that can be specialized in many different ways to provide a variety of specific traffic source models, some of which are described in the next sections.

Example 3.3 A Consider an application that generates messages of M bytes every T seconds. This source can be represented by a MWPP with parameters $\mathcal{A} = \{T\}$, $\mathcal{B} = \{1\}$, $\mathcal{W} = \{M\}$, since arrivals occur regularly with period T, at each arrival corresponds one single message, and the message size is M.

Suppose now that each message needs to be split in K packets to be forwarded across a certain data link layer. Then, from the perspective of the data link layer, the packet generation model is a MWPP with $\mathcal{A} = \{T\}$, $\mathcal{B} = \{K\}$, $\mathcal{W} = \{M/K\}$.

3.3.3 Renewal reward processes

A specific case of a PP is the Renewal Process, which requires the inter-arrival times A_i to be i.i.d. r.v.s with a common statistical distribution $p_{\mathcal{A}}(\cdot)$ (see Sec. **??**). Each arrival

[1] Typically, $B_i \geq 1$. However, it is sometimes convenient to consider $B_i \geq 0$, with $B_i = 0$ indicating events that do not bring any workload to the system. This "silent" event may correspond, e.g., to the generation of a packet that is dropped before transmission.

represents a *renewal time* of the process, as the statistical evolution of the process no longer depends on the past, but only on the nominal distribution of the inter-arrival times.[2]

When the distribution of the inter-arrival times is exponential with parameter λ, then the Renewal process is a Poisson Point Process (PPP) with parameter λ (see Sec. **??**). Accordingly, the number of arrivals in a time interval $(t_1, t_2]$ is a r.v. with Poisson distribution of parameters $\lambda(t_2 - t_1)$, and the arrivals in disjoint time intervals are independent.

Renewal-reward processes (see Sec. **??**) extend Renewal processes by associating *rewards* $\{R_n\}$ with events. In the context of traffic modeling, the reward can be bound to the batch size, or the workload, or both. The *elementary renewal theorem* provides a foundational result for analyzing the long-term behavior of such processes. If rewards are i.i.d. with expected value $\mathrm{E}[R]$, then the expected reward rate is given by:

$$\lim_{t\to\infty} \frac{1}{t} \sum_{n=1}^{N(t)} R_n = \frac{\mathrm{E}[R]}{\mathrm{E}[A]} .$$

This theorem links the long-term average reward to the expected reward per renewal and the expected renewal time.

Example 3.3 B Consider again the Example 3.3 B. Assuming $\mathcal{B}$ and $\mathcal{W}$ are independent of $\mathcal{A}$, the average rate of the traffic generated by the source can be expressed as the ratio between the average workload generated in a renewal interval and the mean duration of a renewal interval, i.e.,

$$\mathrm{G} = \frac{\mathrm{E}\left[\sum_{k=1}^{B_i} W_{k,i}\right]}{\mathrm{E}[A_j]} = \frac{\mathrm{E}[B_\ell]\,\mathrm{E}[W_{k,i}]}{\mathrm{E}[A_j]} . \tag{3.3}$$

3.4 Classification of Source Traffic Models

The behavior of traffic generated by sources in communication networks can be classified into several distinct models, each reflecting specific characteristics of data generation and transmission. These models are critical for understanding and analyzing network performance, as they help capture the diversity of traffic patterns exhibited by different applications. In this section we list the main models, which represent a first high-level classification of traffic generation patterns. This classification provides a foundation for selecting appropriate models based on the nature of the application or network scenario under consideration.

It is important to remark that the classification of a traffic source depends on the behavior of the traffic rate that, in turn, is defined with respect to a certain time interval. By changing the reference time interval, the classification of the traffic source can also change. For example, the traffic generated by a node that sends packets of slightly variable size at regular time intervals can be seen as continuous byte stream of constant rate if observed at time intervals much larger than the inter-packet time, while at a closer look, the traffic pattern appears more variable and dynamic.

[2]It should be noted that this condition is generally not satisfied at time instants other than the renewal times, unless the inter-arrival times are exponentially distributed.

3.4.1 Saturated sources

The saturated source is a theoretical model that is commonly used to test the performance of a system under heavy traffic conditions. Indeed, a *saturated source* assumes the source is always *backlogged*, i.e., with new packets ready for transmission as soon as permitted by the resource manager.

Mathematically, the model can be described as a MWPP with very small inter-arrival times $\{A_i \simeq 0\}$, unit marks $\{B_i = 1\}$, and constant or variable workload. Since in simulations it is not possible to generate zero inter-arrival times (which would require the generation of an infinite number of simultaneous arrivals), the saturated traffic source is typically simulated by generating a new packet as soon as the previous packet leaves the network interface, so as to always have a pending packet ready for transmission.

3.4.2 Constant Bit Rate

The CBR model is used to represent sources that generate constant-size packets periodically. Mathematically, a CBR source can be represented by a MWPP with $\mathcal{A} = \{A\}$, $\mathcal{B} = \{B\}$, $\mathcal{W} = \{W\}$, where A is the (constant) inter-arrival time, B is the (constant) number of packets generated at each arrival event, and W is the (constant) packet size.

This is the case, for example, of the traffic generated by a digital voice call coded with the G.711 coding scheme. In this case, the voice signal is sampled at 8 kHz, and each sample is quantized in 256 levels and represented with 8 bits, so that the output traffic from the voice coder is CBR with an average rate of 64 kbit/s. To be carried across the Internet, the bitstream is split into packets, whose (constant) size dependeds on the packetization interval. For G.711, the most common intervals are 10 ms, 20 ms, or 30 ms, corresponding respectively to packets of 80 bytes, 160 bytes, or 240 bytes. To be noticed that the actual packets at network layer are longer because of the control information (headers) added by the other protocols, namely the Real Time Protocol (12 bytes of header), UDP (8 bytes of header) and IP (20 bytes of header).

3.4.3 Variable Bit Rate

VBR models are widely used in case of multimedia sources, where the data rate is inherently tied to the characteristics of the content being transmitted. For example, in video streaming, the datarate required to transmit a sequence of frames encoded with some advanced video-compression technique can vary substantially from scene to scene, depending on the level of motion, the complexity of the scene, and the compression algorithm used. A static scene with minimal motion can be typically represented with fewer bits, while a fast-moving or highly detailed scene demands a significantly higher data rate. Such variability is efficiently captured using VBR models, which can also reflect real-world constraints such as bandwidth limitations or processing power at the source.

Beyond multimedia applications, VBR sources are also relevant in representing traffic from interactive or event-driven systems, such as online gaming, video conferencing, or Internet of Things (IoT) networks. In these contexts, the traffic generated often depends on external stimuli, such as user actions, environmental changes, or network feedback. For instance, an IoT device monitoring environmental conditions may transmit small amounts of data at regular intervals but produce larger bursts of traffic in response to detected anomalies. Similarly, in a video conferencing scenario, the traffic rate may increase during peri-

ods of active speech or high motion in the video feed and decrease during silence or low activity.

The mathematical modeling of VBR sources typically involves stochastic processes that can capture their dynamic behavior. The MWPP modeling of a VBR source requires at least one of the processes $\mathcal{A}$, $\mathcal{B}$ and $\mathcal{W}$ to be variable. Techniques such as Markov-Modulated Poisson Process (MMPP) (see later) are commonly employed to represent the variability in arrival rates and packet sizes. Additionally, advanced techniques like fractional Brownian motion and heavy-tailed distributions can be applied to model the self-similar and long-range dependent properties often observed in VBR traffic.

3.4.4 Bursty Sources

Bursty sources are a class of traffic models that exhibit alternating periods of high and low activity, often characterized by abrupt changes in transmission rates. This behavior, referred to as *burstiness*, is a fundamental property observed in many real-world applications. Bursty traffic patterns are inherently irregular and unpredictable, making them challenging to model and analyze, yet they play a crucial role in understanding and designing communication networks.

The bursty nature of these sources stems from the interaction between user behavior, application requirements, and the underlying protocols. For example, in web browsing, users typically alternate between downloading content, such as HTML pages, images, or videos, and periods of inactivity while consuming the downloaded material. During active periods, large volumes of data may be transmitted in quick succession, followed by intervals of little or no traffic. Similar patterns are observed in other applications, such as online gaming, where bursts of data are generated in response to player actions, or file transfer protocols, which may transmit large files in segmented bursts.

Quantifying burstiness is essential for accurately characterizing traffic patterns and assessing their impact on network performance. Several metrics are used to describe the degree of burstiness in a traffic source. One common measure is the *peak-to-mean ratio*, which compares the maximum transmission rate observed during a burst to the average rate over time. A higher ratio indicates more pronounced bursts.

Another important metric is the *coefficient of variation* (CV) of interarrival times, defined as the ratio of the standard deviation to the mean. A CV greater than one typically indicates significant variability in arrival times, which is a hallmark of bursty traffic.

Additionally, the *index of dispersion* (also known as the Fano factor) measures the variance-to-mean ratio of the number of arrivals in a given time interval, providing further insight into the temporal variability of traffic.

Mathematical models of bursty sources often employ stochastic processes to represent alternating periods of high and low activity, as explained in the following sections.

3.4.5 On-Off model

The On-Off model is one of the simplest representations of bursty traffic. In this model, the source alternates between two states: an "ON" state, during which CBR or VBR traffic is actively generated, and an "OFF" state, during which no traffic is produced.

The On-Off source can be modelled as a stochastic process with two states, namely ON and OFF. The time spent in a state, before moving to the other, is named *sojourn time*, and is typically assumed to be random. Then, let t_{ON} and t_{OFF} be the r.v.s representing the sojourn times in the ON and OFF states, respectively. Typically, the sojourn times

are assumed to be i.i.d. for each state, and usually independent between states. We indicate their Cumulative Distribution Functions (CDFs) as $F_{ON}(t)$ and $F_{OFF}(t)$, with mean $T_{\text{ON}} = \mathrm{E}[t_{\text{ON}}]$ and $T_{\text{OFF}} = \mathrm{E}[t_{\text{OFF}}]$, respectively. Common probability distributions for these r.v.s are exponential, Pareto, or heavy-tailed. Extensions of the On-Off process, such as the MMPP (see later), allow for more complex and realistic representations of bursty traffic.

We observe that an On-Off cycle represents a renewal interval of duration $t_{\text{Cycle}} = t_{\text{ON}} + t_{\text{OFF}}$ and mean $T_{\text{Cycle}} = T_{\text{ON}} + T_{\text{OFF}}$. If traffic is generated only during the ON period, we can compute the peak rate of the source (that is, the traffic generation rate during the active phase) by analizing the mean traffic of a modified process, simply obtained by removing the OFF periods. The resulting modified process will therefore always be in the active phase. Therefore, if the packet generation process is described by a MWPP $(\mathcal{A}, \mathcal{B}, \mathcal{W})$, then the peak rate (i.e., , the traffic rate in the ON phase) is given by

$$\mathrm{G}_{\text{ON}} = \frac{\mathrm{E}[B_\ell]\,\mathrm{E}[W_{i,k}]}{\mathrm{E}[A_j]}\,. \tag{3.4}$$

The overall average traffic rate generated by the On-Off source, instead, can be computed by scaling (3.4) by the probability that the source is in the ON state. Since each On-Off cycle represents a renewal event, we can model the number of bits generated in a renewal interval as the reward gained during that renewal cycle and then apply the renewal reward theorem. This gives:

$$\mathrm{G} = \frac{\mathrm{E}\left[\sum_{i=1}^{N(t_{\text{ON}})} \sum_{k=1}^{B_i} W_{k,i}\right]}{\mathrm{E}[t_{\text{Cycle}}]} = \frac{\frac{T_{\text{ON}}}{\mathrm{E}[A_j]}\mathrm{E}[B_\ell]\,\mathrm{E}[W_{i,k}]}{T_{\text{ON}} + T_{\text{OFF}}} = \frac{\mathrm{G}_{\text{ON}} T_{\text{ON}}}{T_{\text{ON}} + T_{\text{OFF}}}\,. \tag{3.5}$$

A straightforward extension of the On-Off model assumes that some traffic can also be generated during the Off state, at an average rate $\mathrm{G}_{\text{OFF}} < \mathrm{G}_{\text{ON}}$. In this case, the average traffic rate generated by the source is given by

$$\mathrm{G} = \frac{\mathrm{G}_{\text{ON}} T_{\text{ON}} + \mathrm{G}_{\text{OFF}} T_{\text{OFF}}}{T_{\text{ON}} + T_{\text{OFF}}}\,, \tag{3.6}$$

as can be readily derived using the same reasoning as above.

3.5 Markovian Traffic Source Models

Markovian traffic source models are a class of stochastic models widely used to represent traffic behavior in communication networks. These models rely on the mathematical framework of Markov processes, in which the future state of the system depends only on its current state. This property makes them particularly suitable for describing traffic whose evolution can be represented through a finite or countable set of operational states. Markovian models can capture a variety of traffic characteristics, including variability, burstiness, and transitions between different activity states.

In general terms, a Markovian traffic source model can be described by a Markov chain, that is, a discrete-time Markov process ω_n taking values in a discrete state space $\Omega \subseteq \mathbb{Z}$. Each state is associated with a different operational mode of the traffic source (e.g., idle, active, or bursty), corresponding to a particular traffic-generation pattern. Transitions between states represent changes in the operational mode of the source, such as a transition from an idle state to an active state.